Abdelrahman Khalifa

Software Engineer | Rust, C++, Linux

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SUMMARY

Software engineer working in Rust, modern C++, C and Python. Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting): built the Printers page for COSMIC Settings, with 11 PRs merged upstream. At EKSON, builds the motion simulators' firmware, desktop app and backend. Codeforces Expert, 4-time ECPC finalist.

EXPERIENCE

Google Summer of Code Contributor | **The Linux Foundation (OpenPrinting)** | May 2026 – Oct 2026

Open-source project: Printers page for COSMIC Settings, in Rust; passed the final evaluation | *github.com/Abd002/GSOC-2026*

- Built the **Printers page for COSMIC Settings** and a standalone app with the same UI: printers grouped by device, status and supply levels, print queues, live updates, and an Add Printer flow for legacy printers (cosmic-settings #2182, in review).
- Designed it as 5 Rust crates and ran the C FFI backend as a varlink IPC service in the settings daemon, so a native crash cannot take down Settings.
- **11 PRs merged upstream** (list): ported cups-rs to libcups3 with CUPS 2 and 3 builds and CI, added DNS-SD bindings, and fixed crashes and a leaked client in libcups (C).

Embedded Software Engineer | **EKSON** | Aug 2025 – Present

Software for 3-DoF and 6-DoF pneumatic motion simulators: firmware, desktop app, backend and CI

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.
- Developed the PyQt5 control app; used asyncio to wait for ESP32 and server messages concurrently, cutting CPU use and latency by about 40% (measured).
- Engineered remote operation and firmware updates: a Node.js/Express + MongoDB backend that the desktop app polls over REST to relay commands and flash new firmware with esptool; GitHub Actions CI.

Embedded Linux Systems Intern | **Orange Digital Center (ODC)** | Dec 2024 – Jan 2025

- Created custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing. Scored 100% in the final evaluation.

OPEN SOURCE PROJECTS

Contributor (all merged) — Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 in 6 networking modules (#86977).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214).

PROJECTS

- **Database Systems, CMU 15-445** (C++17): thread-safe Count-Min Sketch for BusTub (per-row locks, merge, top-k); SQL in DuckDB. GitHub
- **Kernel Monitor** (C, Linux, ARM): character device driver exporting per-task memory, threads and CPU time (RCU). GitHub
- **Operating System Simulator** (Java, team of 3): built the kernel, system calls and processes (round-robin, mutexes). GitHub
- **Inventory Management System** (C#, WinForms, SQL): stock and orders app with role-based access. GitHub
- **Team projects:** Sarh Al-Tafawuq web platform (live); Nafaqati, a Flutter expense tracker (Google Play).

TECHNICAL SKILLS

Languages: Rust, Modern C++, C, Python (asyncio), JavaScript (Node.js), SQL, Java, C#, Bash

Computer science: Data structures and algorithms, operating systems, concurrency and multithreading, async/event-driven design, memory management, networking (TCP/UDP, IPC)

Systems & Linux: Linux (Ubuntu/Debian), FFI, IPC (varlink), CUPS/IPP, DNS-SD/mDNS, kernel modules and device drivers, Zephyr RTOS, FreeRTOS, Yocto, Buildroot

Backend & tools: Node.js, Express, MongoDB, REST APIs, PyQt5; Git/GitHub, GitHub Actions CI, Cargo, CMake, testing, debugging, code review

ACHIEVEMENTS

- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024 (Egyptian Collegiate Programming Contest).
- Treasurer, IEEE Student Branch, Minia University: managed a budget of more than $10k.

EDUCATION

Bachelor of Science (B.Sc.), Computer and Systems Engineering | **Minia University** | 2020 – 2025

- Grade: Very Good with Honors.